

Adaptive Multi-Resolution Grid-Based Clustering for Massive Data Streams

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Abstract

The rapid growth of streaming data generated by sensors, Internet of Things devices, online platforms, and scientific applications has created a need for clustering algorithms that can process data online under strict memory constraints. Grid-based density clustering methods, such as D-Stream, are attractive for data stream clustering because they summarize incoming points using grid cells instead of storing the entire stream. However, fixed-resolution grid methods suffer from two major limitations: their memory consumption grows rapidly with dimensionality, and a single global cell size may be unable to capture clusters with different local densities or evolving structures.

In this paper, we propose an adaptive multi-resolution grid-based framework for clustering massive data streams. The method maintains a hierarchical grid structure in which each cell stores compact statistical summaries, including time-decayed density, center of mass, variance, and update time. Cells are dynamically refined when their current resolution is insufficient to represent the local data distribution, and they are merged when high resolution is unnecessary or memory pressure increases. Sparse cells are periodically pruned to reduce memory usage. Clusters are extracted from dense connected grid regions using a density-based procedure inspired by DBSCAN.

The proposed framework improves the memory-accuracy trade-off of fixed-resolution grid-based stream clustering algorithms and provides a flexible mechanism for handling non-uniform data distributions and concept drift.

Keywords: Data stream clustering; massive data; density-based clustering; grid-based clustering; multi-resolution grid; D-Stream; concept drift; memory-efficient algorithms.

1 Introduction

Massive data streams arise naturally in many modern applications, including sensor networks, Internet of Things systems, financial transactions, web logs, network monitoring, and computational biology. In such settings, data points arrive continuously and potentially without bound. Therefore, traditional clustering algorithms, which assume that the entire dataset is stored in memory and can be processed multiple times, are often infeasible.

The data stream model imposes two important constraints. First, the algorithm must process each data point online, usually in one pass. Second, the available memory is significantly smaller than the size of the stream. As a result, stream clustering algorithms typically maintain a compact summary of the observed data rather than storing all incoming points.

Clustering is one of the fundamental tasks in unsupervised learning and data mining. Its goal is to group similar objects together while separating dissimilar objects. Among different clustering approaches, density-based methods are especially useful because they can detect clusters of arbitrary shape and distinguish noise or outliers from meaningful dense regions. However, applying density-based clustering directly to data streams is challenging because density estimates must be updated continuously and efficiently.

Grid-based density clustering methods address this challenge by partitioning the data space into cells and maintaining density summaries for these cells. A well-known example is D-Stream, which uses a fixed-resolution grid and a fading function to give more importance to recent data. Although such methods are efficient, they suffer from important limitations. The number of cells grows exponentially with the dimension of the data, and a fixed grid resolution may be inappropriate for data whose density varies across space or changes over time.

To address these limitations, this paper proposes an adaptive multi-resolution grid-based clustering framework for massive data streams. Instead of using a single global grid resolution, the proposed method dynamically refines or merges grid cells according to local statistical properties of the stream and the available memory budget. Dense or complex regions can be represented by smaller cells, while sparse or simple regions can be summarized by larger cells.

The main idea is to maintain a hierarchical grid structure, such as a quadtree in two dimensions or a 2^d -tree in d dimensions. Each cell stores a compact feature vector describing the local data distribution. Based on this vector, the algorithm decides whether the cell should be refined, merged, pruned, or kept unchanged.

1.1 Contributions

The main contributions of this paper are as follows:

- We propose an adaptive multi-resolution grid framework for density-based clustering of massive data streams.
- We introduce dynamic refinement and contraction operations that allow the grid resolution to change over both space and time.
- We maintain compact cell-level summaries, including time-decayed density, center of mass, variance, and update time, to guide refinement and merging decisions.
- We incorporate sparse-cell pruning to reduce memory consumption and remove cells that have negligible influence on clustering quality.
- We describe a density-based clustering procedure over active grid cells, reducing the need to cluster raw data points.
- We provide a framework that can be viewed as an adaptive multi-resolution extension of fixed-grid stream clustering algorithms such as D-Stream.

2 Related Work

Data stream clustering has been widely studied due to the increasing importance of real-time data analysis. Existing methods can be broadly categorized into micro-cluster-based methods, density-based methods, grid-based methods, and hybrid approaches.

CluStream is one of the early frameworks for clustering data streams. It separates the process into an online micro-clustering phase and an offline macro-clustering phase. DenStream extends the micro-cluster idea using density-based potential and outlier micro-clusters. DBSTREAM also follows a density-based strategy and maintains shared density information between micro-clusters.

Grid-based methods partition the data space into cells and maintain summaries for these cells. STING is an early statistical grid-based method that stores statistical information in hierarchical grid cells. D-Stream adapts grid-based clustering to data streams by using a density decay factor and periodically removing sparse grids. These methods are efficient, but their performance depends heavily on the choice of grid resolution.

The main weakness of fixed-grid approaches is that one global cell size cannot represent all regions of a non-uniform or evolving data stream equally well. If the cell size is too large, distinct clusters may be merged. If the cell size is too small, the number of cells may become too large, causing high memory consumption and computational overhead.

Multi-resolution and adaptive-grid approaches. The idea of letting resolution vary across a stream rather than fixing it once is not new. HASTREAM [9] builds a hierarchical density-based model over micro-clusters and adapts its density threshold independently per region without user intervention, closest in spirit to the present work’s goal, though its mechanism is a hierarchy over micro-clusters rather than a split/merge grid, and it does not maintain the explicit per-cell moment statistics this paper’s refinement criterion is built on. Earlier adaptive-grid proposals include mesh-refinement-style grids that apply different granularities to regions of different density, and accuracy-guaranteed adaptive-grid k -median clustering; both predate the time-decayed density model and the split/merge/prune loop used here, and neither reports the memory-accounting or dense-threshold interactions identified in Section 8.3. To our knowledge, no prior grid-based method couples split/merge decisions directly to a per-cell equidistribution test of the kind studied in Section 8.6 below.

Non-convex and arbitrary-shape streaming clusters. CEDAS [7] and, more recently, MCMSTStream [13] address arbitrary cluster shape directly, the latter by running a minimum spanning tree over KD-tree-indexed micro-clusters rather than axis-aligned cells. This is a relevant contrast to the present work’s own moons-drift result (Section 8.7), where axis-aligned rectangular cells and face-adjacency connectivity are shown empirically to be a poor fit for non-convex boundaries at any grid resolution; combining a micro-cluster-style shape mechanism with the memory accounting of a grid is a natural extension, noted in Future Work.

Concept drift beyond time-decay. This paper follows D-Stream’s exponential fading for drift adaptation. An alternative recent direction augments grid-density clustering with explicit drift detectors: Rezaei and Sajedi [14] combine a grid-density-based clustering model with dual drift-detection mechanisms based on distance metrics and fractal dimension, triggering re-partitioning rather than relying on decay alone. Explicit drift detection of this kind is listed as a future-work direction in this paper as well (see Future Work below), independent of, and potentially complementary to, the density-normalization fix identified as the current bottleneck in Section 8.3.

Where grid-based methods stand in recent surveys. Broader stream-clustering surveys and evaluations, from Mansalis et al. [10] and Ghesmoune et al. [11] through Tareq et al.’s systematic review specifically of density-grid-based methods [12] and the most recent evaluation survey by Nordahl et al. [15], consistently report that density-grid-based algorithms trail micro-cluster-based methods (DenStream-, DBSTREAM-, and CluStream-style approaches) on cluster quality for evolving streams, and identify this as an open, recurring issue across the density-grid literature rather than a property of any single implementation. This is consistent with, rather than contradicted by, the empirical ordering found throughout Section 8 of this paper (imported `river` baselines outperforming both the fixed-grid frontier and `AdaptiveDStream` on ARI at a fraction of the memory): it situates the gap identified in Section 8.3 as an instance of

a wider, already-documented pattern in the density-grid family, not an anomaly specific to this prototype.

Our method is motivated by these limitations. It combines the compact representation of grid-based methods, the time-decayed density mechanism of D-Stream, the statistical summaries of STING-like methods, and the sparse-cell pruning philosophy of stream clustering algorithms such as CEDAS.

3 Problem Definition

Let

$$X = \{x_1, x_2, \dots, x_t, \dots\}$$

be a data stream, where each point $x_t \in \mathbb{R}^d$ arrives at time t . The stream may be very large or potentially infinite, so the algorithm cannot store all points.

The goal is to maintain, at any time t , a clustering of the points observed so far, or more precisely, a clustering of their compact summaries. The algorithm should satisfy the following requirements:

- It should process each point online.
- It should use memory significantly smaller than the number of observed points.
- It should adapt to changes in the data distribution.
- It should detect clusters of arbitrary shape.
- It should handle noise and sparse regions efficiently.

3.1 Time-Decayed Density

Following D-Stream, we assign higher importance to recent data points. Let $\lambda \in (0, 1)$ be a decay factor. If a point x_i arrived at time $T(x_i)$, its density contribution at time t is defined as

$$D(x_i, t) = \lambda^{t-T(x_i)}.$$

For a grid cell g , the time-decayed density at time t is

$$D(g, t) = \sum_{x_i \in E(g, t)} \lambda^{t-T(x_i)},$$

where $E(g, t)$ is the set of points assigned to cell g before or at time t .

If the last update time of cell g is t_l and a new point arrives at time t_n , the density can be updated efficiently by

$$D(g, t_n) = \lambda^{t_n-t_l} D(g, t_l) + 1.$$

This update requires constant time for the affected cell.

4 Proposed Method

The proposed method maintains an adaptive hierarchical grid. Unlike fixed-resolution grid methods, different regions of the data space may be represented at different resolutions.

4.1 Hierarchical Grid Structure

The data space is initially represented by a coarse grid. Each grid cell may be recursively divided into smaller child cells. In two dimensions, this structure corresponds to a quadtree. In d dimensions, each refinement divides a cell into 2^d child cells.

Each cell g has a level $\ell(g)$, where lower levels correspond to coarser cells and higher levels correspond to finer cells. The side length of a cell at level ℓ is denoted by h_ℓ .

4.2 Cell Summary

Each active cell stores a compact summary vector:

$$\Phi(g) = (D(g), \mu(g), \sigma^2(g), t_g, \ell(g), n(g)),$$

where:

- $D(g)$ is the time-decayed density of the cell,
- $\mu(g)$ is the time-decayed center of mass,
- $\sigma^2(g)$ is the estimated variance,
- t_g is the last update time,
- $\ell(g)$ is the cell level,
- $n(g)$ is the number of points or weighted count assigned to the cell.

The center of mass can be updated online. If a new point x_t is assigned to cell g , then after applying decay to the previous statistics, the updated center can be computed as

$$\mu(g, t) = \frac{\lambda^{t-t_g} D(g, t_g) \mu(g, t_g) + x_t}{\lambda^{t-t_g} D(g, t_g) + 1}.$$

Similar online updates can be used for the variance.

4.3 Refinement Criterion

A cell should be refined when its current resolution is not sufficient to represent the local data distribution. We define a refinement score:

$$R(g) = \alpha_1 \cdot \frac{\sigma^2(g)}{h_{\ell(g)}^2} + \alpha_2 \cdot \frac{\|\mu(g) - c(g)\|^2}{h_{\ell(g)}^2} + \alpha_3 \cdot \rho(g),$$

where:

- $c(g)$ is the geometric center of the cell,
- $h_{\ell(g)}$ is the side length of the cell,
- $\rho(g)$ is an optional local density irregularity measure,
- $\alpha_1, \alpha_2, \alpha_3$ are weighting parameters.

The intuition is as follows. If the variance inside a cell is large, or if the center of mass is far from the geometric center, then the assumption that data is uniformly distributed inside the cell may be inaccurate. In such cases, the cell may need to be refined.

A cell g is refined if

$$R(g) > \tau_{\text{split}},$$

provided that the memory budget allows refinement and the maximum grid depth has not been reached.

4.4 Alternative Refinement Criteria: Weyl and Random-Projection Tests

$R(g)$ is a moment-matching statistic: it compares the cell’s tracked mean and variance to the values they would take under a uniform distribution, and is blind, by construction, to any alternative that preserves those two moments while changing the shape of the distribution (e.g. points concentrated along a diagonal of the cell, or a symmetric bimodal split). This subsection introduces two criteria that test equidistribution more directly, both compatible with the same $O(1)$ -per-point streaming update discipline as $R(g)$, and both with a threshold τ that is derived from an explicit probabilistic guarantee rather than tuned as a free constant. Throughout, let $u = (x - \ell(g))/h_{\ell(g)} \in [0, 1]^d$ denote a point rescaled into the cell’s own local unit cube, so that H_0 (“ g is locally equidistributed”) is exactly $u \sim \text{Uniform}([0, 1]^d)$.

4.4.1 The Weyl Criterion

For a nonzero integer vector $k \in \mathbb{Z}^d \setminus \{0\}$, define the character $\varphi_k(u) = \exp(2\pi i k \cdot u)$ and its sample mean over the n points currently in g , $S_k = \frac{1}{n} \sum_{j=1}^n \varphi_k(u_j)$. Because the integer characters form an orthonormal basis of $L^2([0, 1]^d)$, under H_0 each φ_k has mean 0 and unit modulus exactly, so $\text{Var}(\text{Re } \varphi_k) = \text{Var}(\text{Im } \varphi_k) = \frac{1}{2}$ with zero covariance, and any two distinct frequencies $k \neq \pm k'$ are exactly uncorrelated (Proposition 1, Appendix A). Fix a frequency set $\mathcal{K} = \{k_1, \dots, k_m\} \subset \mathbb{Z}^d \setminus \{0\}$ with no two frequencies equal or opposite — we use the d axis unit vectors together with the sum and difference of each consecutive coordinate pair, giving $m = 3d - 2$, linear in d — and define the Weyl statistic

$$T_{\text{Weyl}}(g) = \sum_{k \in \mathcal{K}} 2n |S_k|^2.$$

By the multivariate central limit theorem, $T_{\text{Weyl}}(g) \Rightarrow \chi_{2m}^2$ under H_0 as $n \rightarrow \infty$ (Theorem 1, Appendix A). The refinement threshold is therefore not a tuned constant but the $(1 - \alpha)$ quantile of χ_{2m}^2 for a chosen significance level α : g is refined if $T_{\text{Weyl}}(g) > \chi_{2m}^2(1 - \alpha)$. Unlike $R(g)$, which only aggregates per-axis statistics, $\mathcal{K}$ ’s sum/difference frequencies give T_{Weyl} sensitivity to cross-axis (e.g. diagonal) concentration; Section 8.6 measures this directly. The cost is $O(nm) = O(nd)$ per evaluation, versus $O(n \cdot 2^d)$ for a literal 2^d -orthant χ^2 test. A finite frequency set has blind directions — alternatives whose departure from uniformity lies entirely in an untested harmonic — which is a known property of Fourier-type goodness-of-fit tests in the tradition of Neyman’s smooth test [17]; Section 8.6 exhibits one such case.

4.4.2 The Random-Projection Criterion

An alternative to testing specific harmonics of u is to test linear projections of u directly, in the spirit of random-projection goodness-of-fit testing [18]. Fix a unit vector $v \in \mathbb{R}^d$ ($\|v\|_2 = 1$) and let $Y = v \cdot u$. Under H_0 , Y is a sum of independent scaled uniforms, so $\mathbb{E}[Y] = \frac{1}{2} \sum_j v_j$ and $\text{Var}(Y) = \frac{1}{12} \|v\|_2^2 = \frac{1}{12}$ exactly, independent of v and d , and Y is confined to an interval of width $R(v) = \sum_j |v_j|$ (Proposition 2, Appendix A). Y ’s distribution has no simple closed form for general v (it is a weighted, generalized Irwin–Hall law), so rather than an asymptotic approximation we use a finite-sample concentration bound. Fix m random unit directions $v_1, \dots, v_m$, drawn once and independent of the data. For each v_i we run two elementary tests, both comparing a sample mean of i.i.d. bounded variables to its exact null value via Hoeffding’s inequality [16]: a *mean* sub-test on $Y = v_i \cdot u$ against $\frac{1}{2} \sum_j v_{i,j}$, and a *variance* sub-test on $Z = (Y - \frac{1}{2} \sum_j v_{i,j})^2 \in [0, R(v_i)^2]$ against $\frac{1}{12}$. Running each of the resulting $2m$ elementary tests at level $\alpha/(2m)$ and refining g if any one rejects gives, by a union bound,

$$\Pr(\text{refine } g \mid H_0) \leq \alpha \quad \text{for every finite } n$$

(Corollary 1, Appendix A). This is a strictly stronger guarantee, in one respect, than the Weyl criterion’s: it holds at every n , not only asymptotically. The price is that Hoeffding’s inequality uses only boundedness, not the true variance, so it is loose; and since a random unit vector in $\mathbb{R}^d$ has $\mathbb{E}[|v_j|] = \Theta(1/\sqrt{d})$ per coordinate, its range grows as $R(v) = \Theta(\sqrt{d})$, appearing squared in the bound’s exponent. A fixed true deviation therefore becomes exponentially harder to certify, at fixed n , as d grows — a concrete cost of the non-asymptotic guarantee, measured in Section 8.6. As with the Weyl criterion, m directions cost $O(nm) = O(nd)$ per evaluation for $m = O(d)$, and are drawn once (fixed by a seed derived from d) so the criterion is deterministic and reproducible.

4.5 Contraction Criterion

The algorithm may merge child cells into their parent when high resolution is unnecessary. Let $g_1, \dots, g_{2^d}$ be the children of a parent cell p . These children can be merged if their combined representation is sufficiently close to the parent-level approximation.

A possible contraction condition is

$$\max_i R(g_i) < \tau_{\text{merge}},$$

or

$$\Delta(p) < \epsilon_{\text{merge}},$$

where $\Delta(p)$ measures the information loss caused by replacing the children with their parent.

Contraction is especially useful when memory usage approaches the allowed memory budget.

4.6 Sparse Cell Pruning

Cells with very low decayed density are unlikely to contribute significantly to the final clustering. Therefore, the algorithm periodically removes sparse cells.

A cell g is pruned if

$$D(g, t) < \tau_{\text{sparse}},$$

and it has not received new points for a sufficiently long time.

This pruning mechanism reduces memory usage and allows the algorithm to focus on active dense regions of the stream.

4.7 Clustering over Grid Cells

At query time or after a fixed number of updates, the algorithm extracts clusters from the active grid cells.

Each active cell is labeled as:

- **dense**, if $D(g, t) \geq \tau_{\text{dense}}$,
- **transitional**, if $\tau_{\text{sparse}} \leq D(g, t) < \tau_{\text{dense}}$,
- **sparse**, if $D(g, t) < \tau_{\text{sparse}}$.

A graph is constructed over dense cells. Two cells are connected if they are spatial neighbors and their resolutions are compatible. Clusters are then obtained as connected components of this graph. Transitional cells may be assigned to nearby dense clusters if they satisfy a neighborhood condition.

This procedure is similar in spirit to DBSCAN, but it operates on grid cells rather than raw data points.

Algorithm 1 Adaptive Multi-Resolution Grid Stream Clustering

```
1: Initialize root grid cells.
2: Set decay factor  $\lambda$ , thresholds  $\tau_{\text{split}}$ ,  $\tau_{\text{merge}}$ ,  $\tau_{\text{sparse}}$ , and memory budget  $M$ .
3: for each incoming point  $x_t$  do
4:   Locate the finest active cell  $g$  containing  $x_t$ .
5:   Apply time decay to the statistics of  $g$ .
6:   Update  $D(g)$ ,  $\mu(g)$ ,  $\sigma^2(g)$ , and  $t_g$ .
7:   if  $R(g) > \tau_{\text{split}}$  and memory usage is below  $M$  then
8:     Refine  $g$  into  $2^d$  child cells.
9:     Distribute the summary of  $g$  among its children using the available cell statistics.
10:  end if
11:  if memory usage is close to  $M$  then
12:    Increase refinement strictness.
13:    Attempt contraction of low-information child cells.
14:  end if
15:  if pruning condition is triggered then
16:    Remove cells with  $D(g, t) < \tau_{\text{sparse}}$ .
17:  end if
18: end for
19: Extract clusters from connected dense cells.
```

5 Algorithm

6 Complexity Analysis

Let m be the number of active cells currently stored by the algorithm. Since the algorithm does not store all observed points, its memory complexity is

$$O(m),$$

where m is controlled by the memory budget and pruning mechanism.

For each incoming point, the algorithm must locate the corresponding active cell and update its summary. If the hierarchical grid has maximum depth L , then point insertion requires

$$O(L)$$

time in the worst case. For fixed maximum depth, this becomes constant or logarithmic in the number of active cells depending on the implementation.

The refinement of a cell creates 2^d children. Therefore, the cost of a refinement operation is

$$O(2^d).$$

However, refinement is not performed for every point and is controlled by the refinement threshold and memory budget.

The clustering phase operates over active dense cells. If m_d is the number of dense cells, then constructing the adjacency graph and finding connected components requires approximately

$$O(m_d \cdot \eta),$$

where η is the number of neighboring cells checked for each cell. In fixed dimension, η is usually bounded by a constant.

7 Experimental Evaluation

To evaluate the proposed method, it should be compared against representative stream clustering algorithms, including D-Stream, DenStream, CluStream, DBSTREAM, and CEDAS.

7.1 Datasets

All evaluation in this paper uses synthetic streams, generated so that ground-truth cluster labels are known at every time step and specific structural properties (density mismatch, drift, shape) can be isolated one at a time. Five regimes were used, all two-dimensional, $n=2000$ points, seed 7:

- **static_no_drift**: a tight, dense Gaussian cluster and a broad, sparse Gaussian cluster, present simultaneously, with both centers held fixed for the entire stream. The no-drift control.
- **spatial_drift**: the same tight/broad pair, but both centers now drift across three phases. This is the adversarial case motivating the method: a cell size small enough to resolve the tight cluster is wasteful over the broad cluster’s footprint, while a cell size large enough for the broad cluster swallows the tight cluster whole.
- **density_drift**: centers held fixed (as in **static_no_drift**), but the dense cluster’s standard deviation widens phase by phase ($0.15 \rightarrow 0.15 \rightarrow 1.1$). Isolates a change in local density from any change in position.
- **two_gaussian_drift**: two Gaussian clusters of *equal*, constant spread, drifting across three phases. A non-adversarial control: drift without a density mismatch.
- **moons_drift**: two interleaving, non-convex “moon” clusters, rotating and drifting across three phases. Tests arbitrary cluster shape rather than density or drift.

The tight/broad Gaussian pair and its **drift**/density-schedule parameters are implemented in `make_varying_density_stream`; the symmetric and moon-shaped streams are separate generators. All are pure functions of a random seed, so every run is exactly reproducible.

Section 8.8 extends this with three real streaming-clustering benchmarks, the standard triple in the literature: KDD Cup 99 (network intrusion) [19], Forest CoverType (cartographic) [20], and Statlog Shuttle telemetry [21], used to check whether the synthetic-stream findings above generalize to real, correlated feature spaces rather than only isotropic noise dimensions.

7.2 Evaluation Metrics

Every run below reports, via a prequential (test-then-train) protocol — each point is scored by the model’s current state *before* that model is allowed to update on it:

- Adjusted Rand Index (ARI) and Normalized Mutual Information (NMI),
- clustering purity,
- peak memory, measured from model construction through the full run (so a fixed grid’s upfront cell allocation is counted, not only its online updates),
- throughput (points/second) and fraction of points left unassigned.

Grid/density-based models return no label for a point that falls in a non-dense cell. This is scored, not discarded: every such prediction is mapped to a sentinel label treated as its own predicted cluster, following the standard DBSCAN/DenStream "noise" convention — a model that dumps everything into "unassigned" is penalized as if it put all of that into one large incorrect cluster, rather than being scored only on the points it chose to commit to.

The metrics above weight every point in the stream equally regardless of arrival time. That is not a neutral default: `AdaptiveDStream`, `FixedGridDStream`, and two of the three `river` baselines (DenStream's `decaying_factor`, DBSTREAM's `fading_factor` — only `CluStream`'s windowed snapshots genuinely do not decay) all maintain state that deliberately forgets old data, at three different, uncontrolled rates. A flat aggregate ARI/NMI reflects none of that. We therefore additionally report, for a fixed externally-chosen decay rate λ_{eval} applied identically to every model in a sweep (not read from any individual model's own internal decay parameter, since those are not on a comparable scale to begin with): a decay-weighted purity, in which point i (arrival order) is weighted $\lambda_{\text{eval}}^{n-1-i}$, exactly the rate `GridCell.decay_to` uses internally; and a windowed ARI/NMI, ordinary unmodified `adjusted_rand_score/normalized_mutual_info_score` recomputed over only the most recent $\lceil 1/(1 - \lambda_{\text{eval}}) \rceil$ points, the effective sample size of that decay rate. ARI's chance-correction assumes a specific null distribution over contingency tables that changes under point weights, so rather than risk a subtly incorrect decay-weighted ARI, the windowed version uses trusted, unmodified ARI/NMI math restricted to a recency-focused slice of the stream. Every sweep below uses $\lambda_{\text{eval}} = 0.99$, `AdaptiveDStream`'s own `decay`; Section 8.9 reports what this changes.

Numerical results, including a comparison across the five regimes introduced above, are reported in Section 8.

8 Results

This section reports results from a research prototype of the proposed method, implemented in Python, and compared against a swept fixed-resolution D-Stream baseline and three imported reference methods. Since the previous version of this evaluation, three items proposed but not yet implemented in Section 4 have been added: contraction (children merge back into their parent when none of them individually still need the resolution), a border-assignment rule that attaches transitional cells to an adjacent dense cluster rather than leaving them permanently unlabeled, and two alternative split-mass-redistribution strategies alongside the original. We report the resulting numbers as an honest snapshot of an early-stage implementation rather than a demonstration of a finished method: the prototype is still dominated by the baselines on accuracy, though the new features measurably fixed the two things they targeted.

8.1 Experimental Setup

We use a synthetic two-dimensional stream, `varying_density` (seed 7, $n = 2000$ points¹), constructed so that a single fixed grid resolution provably cannot represent both clusters well: a tight, dense Gaussian cluster ($\sigma = 0.18$) and a broad, sparse Gaussian cluster ($\sigma = 1.6$) are present simultaneously throughout the stream, with both centers drifting across three phases. Models are evaluated prequentially, that is, each incoming point is scored against the model's current state before the model is updated on it.

¹Reduced from $n = 4000$ in the previous version of this evaluation: contraction and the transitional-cell assignment pass both add per-point cost on top of the already-present $O(\text{dense}^2)$ clustering-adjacency check that `predict_point_cluster` pays on every point, not only every maintenance interval, and the harness's `tracemalloc` instrumentation multiplies that further ($\approx 6\times$ overhead observed during recalibration). $n = 4000$ stopped being practical to reproduce on demand; $n = 2000$ keeps the same stream and resolution sweep while remaining tractable. The qualitative pattern below is corroborated independently by the regime sweep (Section 8.7) at its own, separately-reduced sample size.

The reference frontier is a fixed-resolution D-Stream swept over ten grid resolutions, $n_{\text{cells per dim}} \in \{2, 3, 4, 6, 8, 11, 16, 22, 32, 45\}$, tracing out an accuracy-versus-memory curve rather than a single operating point. We additionally import three published stream clustering methods from the `river` library, DenStream, CluStream, and DBSTREAM, without modification, so that the adaptive method is compared against real published baselines and not only against a self-authored fixed-grid ablation. Peak memory is measured with `tracemalloc` from before model construction, so that a fixed grid’s upfront cell allocation is counted rather than only the cost of the online update phase. Points that fall in a non-dense cell with no dense neighbor are assigned a sentinel label rather than being dropped, so that a model which abstains on much of the stream is not rewarded for doing so. Unless noted otherwise, `AdaptiveDStream` below uses contraction ($\tau_{\text{merge}} = 0.01$, merge-eligible only once a split is at least 200 points old) and the default `equal_uniform` split strategy; Section 8.4 compares all three strategies directly.

8.2 Accuracy–Memory Frontier in Two Dimensions

Table 1 reports peak memory, Adjusted Rand Index (ARI), Normalized Mutual Information (NMI), and the fraction of unassigned points for the fixed-grid frontier, the three imported baselines, and the adaptive method. Figure 1 plots ARI and NMI against peak memory for all models.

Table 1: Accuracy–memory frontier on the varying-density stream (2D, $n = 2000$, seed 7). `AdaptiveDStream` is highlighted; it is dominated on ARI by the fixed-grid frontier at comparable memory and by DBSTREAM.

Model	Peak memory (KB)	ARI	NMI	Frac. unassigned
FixedGrid $n=2$	7.4	-0.001	0.014	0.00
FixedGrid $n=3$	11.2	0.001	0.002	0.01
FixedGrid $n=4$	16.4	0.001	0.021	0.01
DBSTREAM	23.2	0.236	0.221	0.00
FixedGrid $n=6$	31.8	0.004	0.042	0.03
DenStream	47.9	0.092	0.202	0.00
FixedGrid $n=8$	53.0	0.064	0.200	0.04
FixedGrid $n=11$	96.2	0.029	0.080	0.10
CluStream	99.6	0.061	0.084	0.00
FixedGrid $n=16$	199.4	0.203	0.205	0.27
FixedGrid $n=22$	378.0	0.442	0.403	0.42
AdaptiveDStream	666.4	0.066	0.078	0.21
FixedGrid $n=32$	798.6	0.692	0.627	0.48
FixedGrid $n=45$	1581.3	0.822	0.739	0.51

At its own memory footprint (666 KB, between the fixed grids at $n=22$ and $n=32$), the fixed-grid frontier reaches ARI 0.44–0.69, versus 0.066 for the adaptive method; DBSTREAM reaches higher ARI still (0.236) using roughly 30 times less memory (23 KB). Relative to the previous version of this evaluation (Table 1, $n = 4000$: 1344 KB, ARI 0.106, 42% unassigned), two things changed and one did not. Peak memory roughly halved in proportion to the smaller sample, and **fraction unassigned dropped from 0.42 to 0.21** — a direct, measurable effect of the transitional-cell border-assignment rule, which no longer leaves every non-dense leaf’s points unlabeled. ARI did not improve: 0.066 here is, if anything, slightly below the earlier run’s 0.106. The same pattern, substantial memory and unassigned-fraction gains with no corresponding ARI gain, repeats independently in the regime sweep (Section 8.7), which is evidence this is a real effect of the new features and not sampling noise from the smaller n .

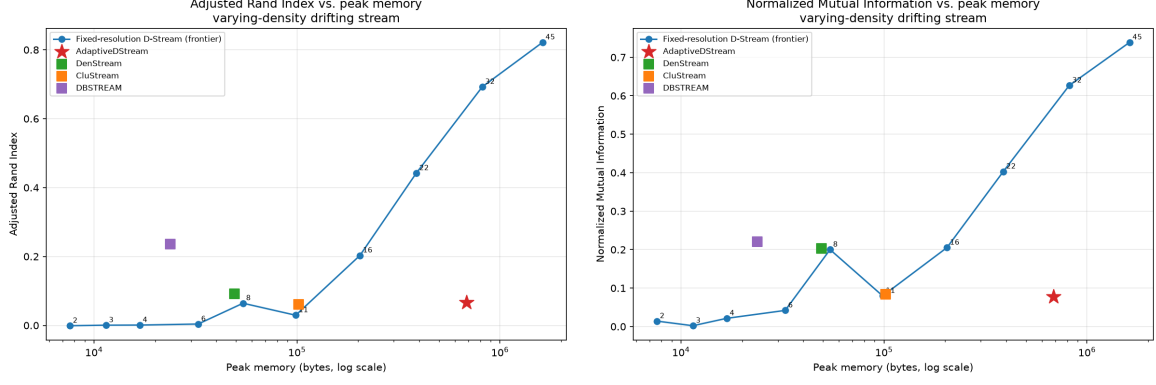


Figure 1: ARI (left) and NMI (right) versus peak memory for the fixed-grid frontier, the three imported baselines, and AdaptiveDStream.

8.3 Root-Cause Analysis of the Accuracy Gap

We traced the original gap to two specific, verified implementation issues, both concerning how the dense/sparse rule and memory accounting interact with splitting, rather than the splitting heuristic (when to split) itself. Contraction and transitional assignment (Section 4) were built to address the second; neither touches the first, and the frontier result above shows exactly that split: the issue they target is measurably better, and ARI is not.

1. **The dense threshold is an absolute decayed count, not a density — unchanged by this iteration.** Splitting a cell fragments its accumulated mass across its 2^d children. A region can be genuinely dense, that is, have high mass per unit volume, while every fine cell covering it holds too little raw decayed count to cross τ_{dense} , simply because the same mass is now divided among more cells. Consequently no single absolute threshold value is correct both before and after a region is refined. We originally verified this is the dominant effect by manually sweeping the threshold pair from $(\tau_{\text{dense}}, \tau_{\text{sparse}}) = (2.0, 0.3)$, the value shared with the fixed-grid baseline, down to $(0.5, 0.05)$: ARI improved from 0.040 to only about 0.106 before plateauing, never approaching the fixed-grid frontier. None of contraction, transitional assignment, or the choice of split strategy (Section 8.4) changes how τ_{dense} and τ_{sparse} are compared against $D(g, t)$, so this remains the top open item.
2. **Splitting still does not fully free the parent cell’s storage — contraction now reclaims some of it, quantifiably.** The split operation attaches 2^d children to a cell but leaves the parent object itself resident in memory, still holding its sufficient statistics, unless the children later satisfy the contraction criterion and are merged back. A direct, instrumented comparison on the same stream and configuration as Table 1: *without* contraction, the run ends with 754 leaf cells and 1005 total live cell objects (251 retired parents, $\approx 25\%$ dead weight, matching the original finding almost exactly); *with* contraction enabled, it ends with 637 leaf cells and 849 total objects (212 retired parents), a real $\approx 15\%$ reduction in total live cells. Contraction does not eliminate the issue, though: a cell whose children keep legitimately deviating from the locally-uniform assumption, that is, a real, persistent cluster, never satisfies the merge criterion, and its stale parent remains resident indefinitely, exactly as before.

Both issues point toward the same fix as before: making the dense/sparse classification resolution-aware. Memory accounting is now partially, though not fully, addressed by contraction.

8.4 Split-Strategy Comparison

A split has to infer how to redistribute a parent’s decayed mass across its 2^d children from summary statistics alone, since raw observations are not kept. Three strategies are implemented (Section 4): `equal_uniform` (the original default: mass split equally, each child assumed locally uniform), `point_mass` (a degenerate baseline: all mass to whichever single child contains the parent’s tracked mean), and `moment_based` (an axis-independent Gaussian model of the parent, truncated to its bounds, giving each child the Gaussian mass and truncated-normal moments its half-interval implies). Table 2 compares all three, with contraction enabled, on the same adversarial stream (`drift=true`, $n = 800$, seed 7), holding every other hyperparameter fixed.

Table 2: Split-strategy comparison, varying-density drifting stream ($n = 800$, seed 7). Only `split_strategy` differs between the three `AdaptiveDStream` rows.

Model	Peak memory (KB)	ARI	NMI	Frac. unassigned
DBSTREAM	22.3	0.019	0.048	0.00
FixedGrid $n=8$	55.7	0.052	0.172	0.08
DenStream	40.2	0.062	0.155	0.00
CluStream	98.9	0.112	0.118	0.00
AdaptiveDStream (<code>equal_uniform</code>)	315.6	0.024	0.084	0.07
AdaptiveDStream (<code>moment_based</code>)	391.4	0.021	0.066	0.10
AdaptiveDStream (<code>point_mass</code>)	447.2	0.017	0.037	0.34
FixedGrid $n=32$	797.8	0.667	0.594	0.53

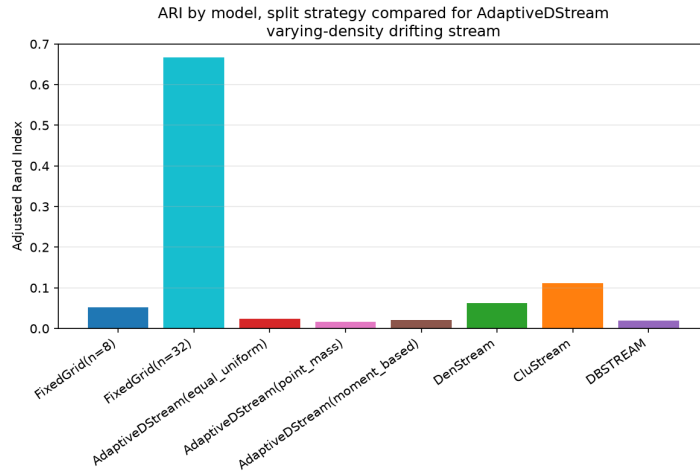


Figure 2: ARI by model on the split-strategy comparison of Table 2.

`equal_uniform`, the simplest strategy, is narrowly the best of the three on this stream, and `point_mass` is clearly the worst, for an explainable reason: it hands each split’s entire mass to one child and leaves the other $2^d - 1$ at exactly zero, so those children stay **sparse** until real new data arrives, inflating fraction unassigned to 0.34 (3–5 $\times$ the other two strategies) while also using the most memory of the three, since an empty child is still a live `GridCell` object. `moment_based` sits in between, as intended: it spreads mass more realistically than `point_mass` (verified separately by an exact mass-conservation test) but still concentrates most of it near the tracked mean, costing it slightly against `equal_uniform`’s even split on this particular stream. None of the three strategies brings `AdaptiveDStream` close to the fixed-grid or **river** baselines: this experiment isolates the split-mass-redistribution question specifically, and the answer is that it is not where the current ARI gap originates, consistent with Section 8.3.

8.5 Effect of Dimensionality

We repeated the comparison on the same stream family generalized to $d > 2$, with the two cluster centers orbiting in the first two coordinates and every additional dimension carrying pure isotropic noise, the standard curse-of-dimensionality stress case. This is a preliminary probe rather than a full multi-dimensional evaluation: it uses a single seed, $n = 400$ points (reduced from 2500 for the same reason as Section 8’s experimental setup, and more acutely here since a split costs $O(2^{\text{dim}})$ regardless of strategy and contraction adds a second full-tree pass every maintenance interval), and neither model’s thresholds were retuned per dimension. At each dimension, the fixed grid’s resolution was chosen so its total cell count lands near the adaptive method’s 2500-cell budget, so the comparison is roughly memory-matched rather than resolution-matched; past $d = 15$ the fixed grid’s own resolution floor of $n_{\text{cells per dim}} = 2$ pushes its cell count itself past $2^{20} \approx 10^6$, at which point it is skipped rather than allocating a baseline whose shared-budget premise no longer holds (see Table 3, $d = 20$). Table 3 and Figure 3 summarize the outcome for $d \in \{2, 3, 4, 5, 6, 8, 10, 12, 15, 20\}$.

Table 3: Effect of dimensionality on peak memory and accuracy, at roughly matched memory budgets (varying-density stream, $n = 400$, seed 7, single run per dimension).

Dim	Model	Cells	Peak memory (KB)	ARI	NMI
2	FixedGrid ($n=50$)	2500	1949.8	0.604	0.574
2	AdaptiveDStream	136	143.0	0.021	0.099
3	FixedGrid ($n=14$)	2744	2221.4	0.471	0.460
3	AdaptiveDStream	225	210.1	0.000	0.038
4	FixedGrid ($n=7$)	2401	2019.8	0.497	0.464
4	AdaptiveDStream	466	418.5	0.000	0.030
5	FixedGrid ($n=5$)	3125	2724.6	0.523	0.466
5	AdaptiveDStream	714	642.5	-0.002	0.034
6	FixedGrid ($n=4$)	4096	3711.0	0.705	0.599
6	AdaptiveDStream	946	868.5	0.065	0.156
8	FixedGrid ($n=3$)	6561	6334.4	0.697	0.618
8	AdaptiveDStream	1531	1499.6	0.239	0.212
10	FixedGrid ($n=2$)	1024	1053.5	0.030	0.159
10	AdaptiveDStream	1024	1272.8	0.047	0.116
12	FixedGrid ($n=2$)	4096	4465.9	0.393	0.405
12	AdaptiveDStream	1	2.5	0.000	0.005
15	FixedGrid ($n=2$)	32768	38833.3	0.029	0.121
15	AdaptiveDStream	1	2.5	-0.000	0.005
20	FixedGrid ($n=2$)	skipped: 2^{20} cells exceeds the shared-budget comparison’s validity			
20	AdaptiveDStream	1	2.6	-0.000	0.005

Two patterns emerge, both consistent with the earlier ($n = 2500$) run but sharper. First, the memory hypothesis behind the method holds more strongly than before: from $d = 2$ to $d = 8$, AdaptiveDStream now uses only 7–24% of the budget-matched fixed grid’s memory (143 KB vs. 1950 KB at $d = 2$; 1500 KB vs. 6334 KB at $d = 8$), a larger relative gap than the earlier run’s 35–87%, since contraction reclaims memory from stale splits on top of the `max_cells` cap that was already present. The fixed grid still cannot be tuned finely to a memory budget at high dimension, for the same reason as before: $n_{\text{cells per dim}}$ only moves in coarse integer steps, so at $d = 8$ the jump from $n = 2$ (256 cells) to $n = 3$ (6561 cells) overshoots the 2500-cell target by $2.6\times$. At $d = 10$ the fixed grid again drops under budget (1054 KB) purely because $n_{\text{cells per dim}}$ bottoms out at 2, and AdaptiveDStream (1273 KB) is again the larger of the two there.

Second, AdaptiveDStream remains dominated in accuracy for $d = 2$ –6, and the gap narrows at $d = 8$ exactly as before (ARI 34% of the fixed grid’s, up from single digits at low dimension)

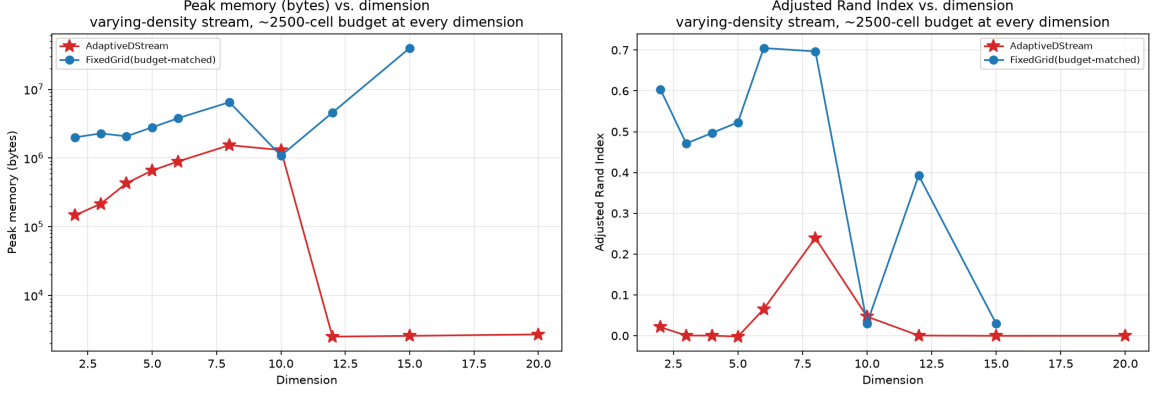


Figure 3: Peak memory (left) and ARI (right) versus dimension, comparing AdaptiveDStream against a fixed grid whose resolution is chosen to match its cell budget at each dimension.

— consistent with the unchanged root cause (Section 8.3): a split fragments mass across 2^d children, a far harsher penalty at $d = 8$ (256-way) than at $d = 2$ (4-way), and the dense threshold remains an absolute count regardless of contraction or transitional assignment. At $d = 10$, AdaptiveDStream (0.047) nominally exceeds the fixed grid (0.030) for the first time — but the fixed grid’s own ARI collapsed sharply relative to the earlier, larger-sample run (0.486 there vs. 0.030 here), most plausibly a small-sample artifact of an $n=2$ -per-axis grid at only 400 points, not evidence that AdaptiveDStream overtook it on merit. Given the single seed, untuned per-dimension thresholds, and the smaller sample size used here, we treat this dimensionality result at $d \leq 10$ as a preliminary signal worth pursuing rather than a validated finding, and read the $d = 10$ reversal specifically as noise.

Third, and the reason this sweep was extended past $d = 10$: at $d = 12$ the root’s admission check for a split (projected leaf count $\leq \text{max_cells} = 2500$) is refused outright, since $2^{12} = 4096$ alone already exceeds the budget before any refinement criterion is even evaluated. AdaptiveDStream is left with exactly one leaf, the whole domain, at $d = 12, 15, 20$ (Table 3), and its ARI collapses to 0 accordingly — a single-cluster prediction against genuinely two-cluster ground truth. This is architectural, not a criterion failure: no refinement statistic, however well calibrated (Section 4.4), can influence a split decision that is vetoed before it runs. Reaching real-data dimensionality (e.g. the ~ 41 features of KDD Cup 99) therefore requires the split rule itself to stop enumerating all 2^d children at once — splitting a single chosen axis, or the top- k axes ranked by per-axis non-uniformity evidence, are the natural candidates — independent of, and prior to, which equidistribution criterion drives the decision. We return to this in the Future Work section below.

8.6 Single-Cell Equidistribution Criteria Comparison

Sections 4.3–4.4 introduce three criteria for the same question — is a cell’s data locally equidistributed? — with different cost/guarantee profiles: $R(g)$ (moments, $O(1)$ streaming, no calibration guarantee), $T_{\text{Weyl}}(g)$ (asymptotically exact chi-square calibration, $O(nd)$), and the random-projection test (exact finite-sample calibration, also $O(nd)$). This experiment isolates that question from the tree/split/merge machinery entirely: each criterion is scored on a batch of $n = 400$ points drawn directly from a known box, with no cell splitting involved, so the comparison is not confounded by the 2^d -split wall of Section 8.5.

Five synthetic shapes were used: **uniform** (H_0), **core_border_imbalance** (a uniform core plus two opposite-face border slabs at different densities from each other), **diagonal_correlated** (points in a thin band around the main diagonal — every 1-D marginal looks close to uniform while the joint distribution is highly concentrated), **symmetric_bimodal** (two Gaussian blobs

symmetric about the box center along one axis, chosen so the mean is exact and the variance is close to the uniform value), and **corner_spike** (a point-mass concentration in one corner plus uniform background). Discriminative power is reported as ROC-AUC (uniform vs. each shape, $R = 60$ repeats), which compares the three criteria without requiring their very differently scaled thresholds to be matched. Figure 4 and Table 4 summarize the results across $d \in \{2, 5, 10, 15, 20\}$ (averaged over dimension in the table; see the figure for the per-dimension trend).

Table 4: ROC-AUC (uniform vs. shape), averaged over $d \in \{2, 5, 10, 15, 20\}$, $n = 400$, $R = 60$ repeats per cell. Bold marks the best criterion per shape.

Shape	$R(g)$ (moment)	T_{Weyl}	Random projection
core_border_imbalance	0.995	0.999	0.543
core_border_imbalance_diag	1.000	1.000	0.903
diagonal_correlated	0.419	1.000	0.508
symmetric_bimodal	0.902	0.477	0.500
corner_spike	1.000	1.000	1.000

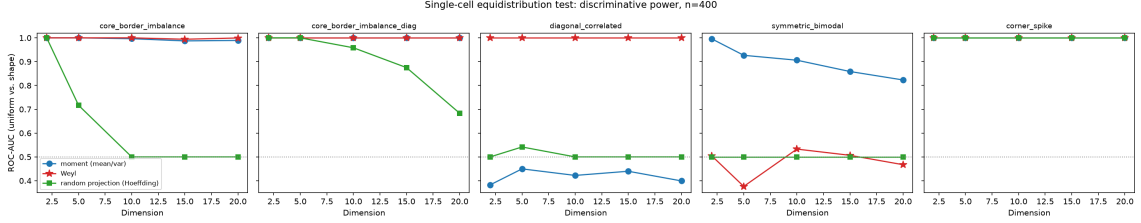


Figure 4: ROC-AUC by shape and dimension for the three refinement criteria. Random projection’s power on the two border-imbalance shapes visibly collapses toward chance as dimension grows, exactly as Theorem 2’s discussion predicts ($R(v) = \Theta(\sqrt{d})$ weakens the bound); Weyl’s power does not.

No criterion dominates. T_{Weyl} is the only one of the three that detects **diagonal_correlated** at any dimension (AUC 1.000 throughout, versus chance-level 0.42–0.54 for both others), the direct payoff of including cross-axis sum/difference frequencies in $\mathcal{K}$ rather than only per-axis statistics. Conversely, $R(g)$ is the best criterion on **symmetric_bimodal**; T_{Weyl} ’s near-chance AUC there (0.38–0.53 across dimension) is not a general weakness but a specific, verified coincidence — the shape’s default separation of 0.25 box-widths from center puts $\cos(2\pi \cdot 0.25) = 0$ exactly, a null harmonic of the $k = 1$ axis frequency, and re-running at separation 0.15 (where $\cos(2\pi \cdot 0.15) \approx 0.59$) restores T_{Weyl} ’s AUC to 1.000 at every dimension tested. This is a known, citable property of Fourier/Neyman-smooth goodness-of-fit tests: a finite frequency set has blind directions, alternatives whose signal lies entirely in an untested harmonic.

The random-projection criterion is the weakest of the three at this sample size, at chance level on every shape except the crude **corner_spike** case, and its already-limited power on the two border-imbalance shapes falls further as dimension grows (Figure 4), from AUC ≈ 1.0 at $d = 2$ to ≈ 0.5 by $d = 10$ –15. This is the direct, measured cost of Corollary 1’s non-asymptotic guarantee discussed in Section 4.4: for a random unit direction in $\mathbb{R}^d$, the Hoeffding range $R(v) = \Theta(\sqrt{d})$ enters squared in the tail bound’s exponent, so a fixed true deviation becomes exponentially harder to certify at fixed n as d grows. It is also structurally blind to **diagonal_correlated** regardless of dimension, since that shape is symmetric about the box center along every direction by construction, so a mean-based sub-test cannot see it and the variance sub-test’s signal along a generic (non-diagonal-aligned) direction is too weak, at $n = 400$, to clear Hoeffding’s conservative bound.

We draw two conclusions for the method as a whole. First, the three criteria have complementary, not overlapping, blind spots, which argues for combining them (e.g. via Fisher’s method on independent p -values) rather than selecting one; we leave this for future work. Second, the false-positive calibration promised in Section 4.4 holds empirically, not only asymptotically: across $n \in \{50, \dots, 6400\}$ at $d = 10$, T_{Weyl} ’s empirical false-positive rate under true H_0 stayed within 0.033–0.070 of its nominal $\alpha = 0.05$ at every n (Figure 5), while the random-projection criterion’s was 0 throughout — strictly within, and well under, its $\leq \alpha$ bound, confirming Corollary 1 is valid and, as its proof anticipates, conservative. $R(g)$ ’s own fixed threshold ($\tau_{\text{split}} = 0.04$) showed 0 false positives in this batch (no-decay) setting at every n tested, but this is not because it is well-calibrated in the same sense: repeating the check with the model’s actual time-decayed statistic (Section 3.1) shows the safety margin depends on the decay factor λ , not on n — at $\lambda = 0.995$ (this paper’s default) no spurious split occurred in 100 runs of 20,000 uniform points, but at $\lambda = 0.8$ every run eventually produced one, since decay caps the effective sample size at $1/(1 - \lambda)$ regardless of stream length. A principled threshold for $R(g)$ would therefore be a function of $1/(1 - \lambda)$, the way T_{Weyl} ’s threshold is a function of (n, m) , rather than a single constant assumed safe at every decay setting.

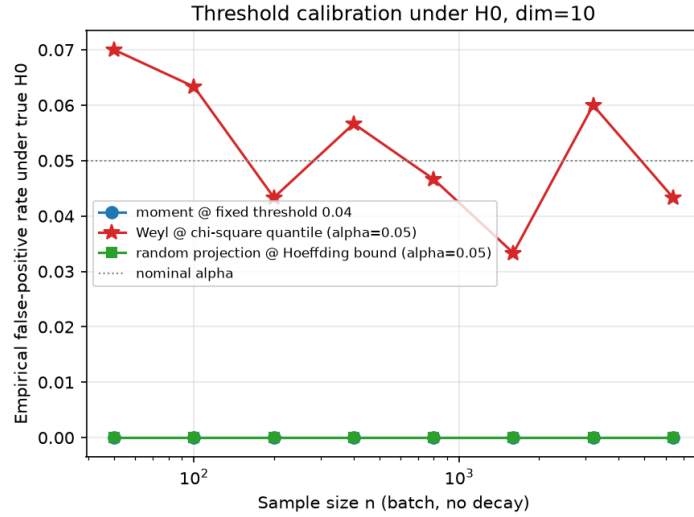


Figure 5: Empirical false-positive rate under true H_0 versus sample size n (batch, no decay), $d = 10$. T_{Weyl} tracks its nominal $\alpha = 0.05$ at every n ; the random-projection criterion stays at or near 0, strictly under its $\leq \alpha$ bound; $R(g)$ ’s fixed threshold shows 0 false positives here but is not calibrated in the same sense (see discussion above).

8.7 Cross-Regime Generalization

The frontier and dimensionality results above answer “accuracy versus memory on one adversarial stream.” This experiment asks a different question: does a single, fixed hyperparameter choice generalize across qualitatively different regimes, or does the right answer depend on what the data is doing? Five two-dimensional regimes were constructed, all $n = 500$ points (reduced from 2000 for the same reason as Section 8’s experimental setup), seed 7: **static_no_drift** (the tight/broad Gaussian pair with both centers frozen, a no-drift control), **spatial_drift** (the same pair with centers drifting across three phases), **density_drift** (centers frozen, the dense cluster’s standard deviation widening phase by phase from 0.15 to 1.1), **two_gaussian_drift** (two equal-spread Gaussian clusters drifting, a non-adversarial control with no density mismatch), and **moons_drift** (two rotating, drifting non-convex clusters). No model’s hyperparameters were retuned per regime: **AdaptiveDStream** (with contraction and transitional assign-

ment) and **FixedGridDStream** at two resolutions ($n_{\text{cells per dim}} \in \{8, 32\}$) keep the configuration from Section 8.3, and the same three **river** baselines are reused, so only the data changes across rows of Table 5.

Table 5: Adjusted Rand Index by model and regime ($n = 500$, seed 7). Bold marks **AdaptiveDStream**, the proposed method.

Regime	FixedGrid $n=8$	FixedGrid $n=32$	AdaptiveDStream	DenStream	CluStream	DBSTREAM
static_no_drift	0.472	0.785	0.123	0.542	0.241	0.543
spatial_drift	0.037	0.764	0.030	0.064	0.133	0.017
density_drift	0.450	0.318	0.109	0.611	0.354	0.422
two_gaussian_drift	0.460	0.008	0.041	0.384	0.340	0.063
moons_drift	0.004	0.001	0.004	0.043	0.116	0.010

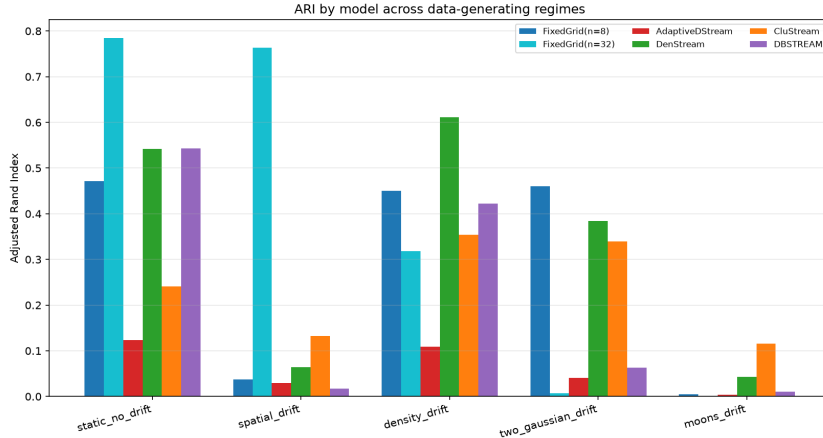


Figure 6: Adjusted Rand Index by model across the five regimes in Table 5.

Peak memory is roughly regime-independent per model: FixedGrid $n=8$ uses 52–57 KB, FixedGrid $n=32$ uses 797–810 KB, **AdaptiveDStream now uses only 183–270 KB** (down from 676–1296 KB before contraction and transitional assignment), DenStream uses 15–37 KB, CluStream uses 95–99 KB, and DBSTREAM uses 17–36 KB. Fraction unassigned for **AdaptiveDStream** is 0.02–0.05 in every regime here, down from 0.34–0.53 previously. Three findings follow.

No single fixed resolution wins across regimes. *spatial_drift* and *two_gaussian_drift* are the clearest contrast: FixedGrid $n=32$ wins *spatial_drift* by a wide margin (0.764 vs. $n=8$ ’s 0.037, since only fine cells can resolve the tight cluster as it moves) and then collapses on *two_gaussian_drift* (0.008 vs. $n=8$ ’s 0.460) — cells fine enough for the adversarial density-mismatch stream are too fine for a stream with no density mismatch to justify them, so a moving symmetric cluster leaves too little decayed mass in any one cell before drifting onward. *density_drift* shows the same reversal from the other side ($n=8$ wins, 0.450 vs. 0.318). This is the empirical case for adaptivity that motivates the method in Section 4 — whichever single resolution is chosen up front, some regime here defeats it.

AdaptiveDStream is dominated in every regime tested, including the static, no-drift control — but contraction and transitional assignment visibly moved the needle without closing the gap. Peak memory and fraction unassigned both improved substantially and consistently across all five regimes (above), exactly what those two features were built to do. ARI itself moved in both directions depending on the regime: up in *static_no_drift* (0.043 $\rightarrow$ 0.123), *density_drift* (0.070 $\rightarrow$ 0.109), and *two_gaussian_drift* (0.006 $\rightarrow$ 0.041); down in *spatial_drift* (0.084 $\rightarrow$ 0.030); flat in *moons_drift*. This is a mixed, modest pattern rather than a clear win, and *static_no_drift* remains far below the best fixed grid or **river** model there (0.123 vs. 0.785/0.543) — consistent with Section 8.3: the non-volume-normalized

dense threshold is untouched by either new feature, so the shortfall remains structural rather than drift-related.

Non-convex shape defeats every model tested, uniformly. On `moons_drift`, no grid-based model exceeds ARI 0.004 and no `river` baseline exceeds 0.116. Axis-aligned rectangular cells, and face-adjacency connectivity between them, are a poor fit for a non-convex cluster boundary at any resolution; this is a shape limitation orthogonal to the density and drift questions the other four regimes isolate, and no amount of resolution tuning among the models compared here would close it.

Among the imported baselines, DenStream and DBSTREAM remain the most consistently competitive without per-regime retuning — DenStream is strongest on `static_no_drift`, `density_drift`, and `two_gaussian_drift`, with DBSTREAM close behind on the first two — each using an order of magnitude or more less memory than either fixed grid or `AdaptiveDStream`.

8.8 Real-Data Memory-Accuracy Frontier

Every result above uses synthetic streams, where ground truth is exact but every dimension beyond the constructed ones is either meaningful signal by design or pure isotropic noise (Section 8.5). This experiment asks whether the same qualitative findings hold on real, correlated feature spaces: KDD Cup 99 (binary normal-vs-attack) [19], Forest CoverType (7-class cartographic) [20], and Statlog Shuttle telemetry (7-class, used as the "sensor" stream) [21]. Each dataset is reduced to its top-6 features by mutual information with the label — a supervised, and therefore not fully realistic, feature-selection step, applied identically to every model so the model-vs-model comparison stays fair — and robust-scaled into $[0, 1]^6$; each stream is systematically subsampled to $\approx 8,000$ – $8,300$ points, preserving coarse temporal/burst structure rather than taking a homogeneous head slice. The same accuracy-memory frontier design as Section 8 is used: `FixedGridDStream` swept over four grid resolutions, `AdaptiveDStream` swept over four `max_cells` budgets ($\{300, 800, 2000, 6000\}$), and the three `river` baselines as single points, all on the identical reduced stream per dataset.

Table 6: Best-of-family Adjusted Rand Index and peak memory, three real datasets, dim= 6. Full sweep (all resolutions/budgets) in `outputs/real_frontier_results.json`.

Dataset	Model	Peak memory	ARI
kddcup99	FixedGrid $n=4$	3708 KB	0.113
kddcup99	AdaptiveDStream (<code>max_cells=6000</code>)	5496 KB	0.132
kddcup99	DenStream	33 KB	0.393
kddcup99	CluStream	192 KB	0.328
covtype	FixedGrid $n=3$	669 KB	0.026
covtype	AdaptiveDStream (<code>max_cells=6000</code>)	5495 KB	0.004
covtype	DBSTREAM	50 KB	0.038
sensor:shuttle	FixedGrid $n=3$	669 KB	0.075
sensor:shuttle	AdaptiveDStream (<code>max_cells=2000</code>)	1800 KB	0.035
sensor:shuttle	DenStream	28 KB	0.423

On real data, neither grid method is competitive with the imported micro-cluster baselines, and `AdaptiveDStream` does not clearly outperform the fixed grid the way its memory advantage on synthetic noise dimensions (Section 8.5) suggested it might. Both `FixedGridDStream` and `AdaptiveDStream` score in a similar, weak, inconsistent range across all three datasets (ARI roughly -0.03 to 0.13), while DenStream and DBSTREAM clear 0.34 – 0.42 using 30 – $190\times$ less memory than `AdaptiveDStream`'s better-scoring configurations, on two of the three datasets (`kddcup99`, `sensor`). `covtype` is the exception

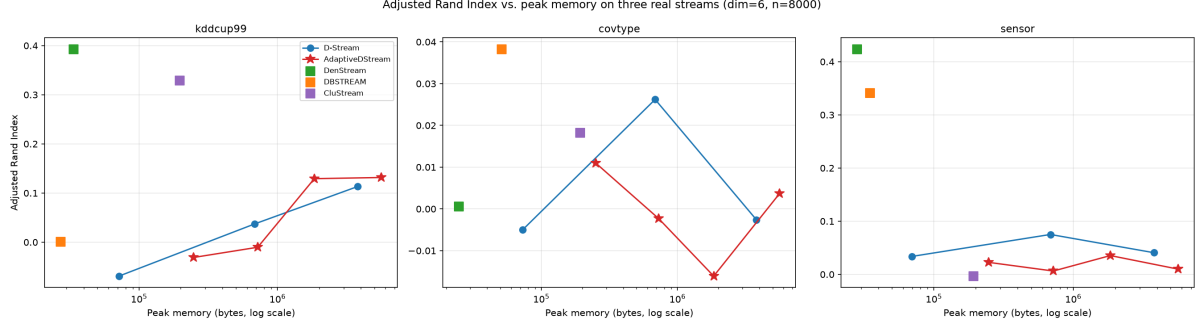


Figure 7: Adjusted Rand Index vs. peak memory (log scale), three real datasets. Circles: FixedGridDStream frontier. Stars: AdaptiveDStream frontier. Squares: river baselines.

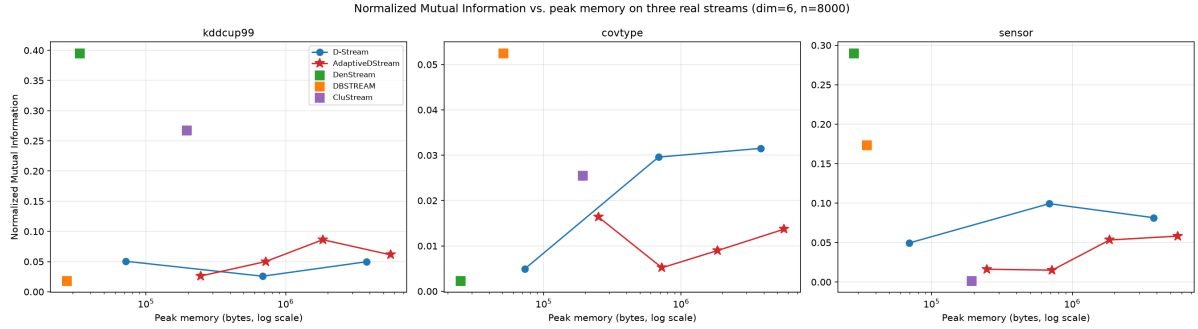


Figure 8: Normalized Mutual Information vs. peak memory (log scale), three real datasets.

where nothing scores well (best ARI anywhere is DBSTREAM’s 0.038), plausibly because the 7-class cartographic problem is not well suited to density-based clustering at all, independent of grid resolution or adaptivity.

A degeneracy check on kddcup99 (`max_cells`= 4000) confirms the 2^d -split-rule wall identified in Section 8.5 on real, not synthetic-noise, features: at $\text{dim}=10$, $2^{10} = 1024 < 4000$ and AdaptiveDStream splits normally, reaching 3,070 leaves (ARI 0.056); at $\text{dim}=12$, $2^{12} = 4096 > 4000$, so `_should_split`’s admission check refuses the very first split and the entire reduced feature space remains one cell (leaves= 1, ARI 0.001) for the whole run, and stays that way at $\text{dim}=15$ and $\text{dim}=20$. This demonstrates the split-rule wall is not an artifact of the synthetic dimension sweep’s isotropic-noise construction: a per-axis, rather than full- 2^d -orthant, split rule is a hard prerequisite for this method to reach realistic feature counts (e.g. kddcup99’s 38 numeric columns) at all, independent of any accuracy-side threshold fix.

8.9 Recency-Weighted Re-Evaluation

Every synthetic sweep above (frontier, split-strategy, dimensionality, cross-regime) was re-run with $\lambda_{\text{eval}} = 0.99$ (Section 8’s Evaluation Metrics), adding the windowed `ari_recent` and decay-weighted `decayed_purity` alongside the flat ARI already reported. This is not a replacement for the earlier numbers: the two lenses tell a genuinely different story depending on the experiment, in both directions, not uniformly a more flattering one for AdaptiveDStream.

Frontier sweep: the gap narrows, but does not close. Table 7 and Figure 9 repeat Table 1 with `ari_recent` added. AdaptiveDStream’s ARI rises from 0.066 (flat) to 0.582 (recent) — but its immediate memory neighbors on the fixed-grid frontier rise further: FixedGrid $n=22$ (378 KB) reaches 0.842, FixedGrid $n=32$ (799 KB) reaches 1.000. DBSTREAM (0.236 $\rightarrow$ 0.481) and DenStream (0.092 $\rightarrow$ 0.397) improve similarly. Recency-weighting helps nearly every model on this stream — a 3-phase drifting stream is hardest for everyone before decayed mass accu-

mulates early on — but the fixed-grid frontier’s dominance over **AdaptiveDStream** survives this lens.

Table 7: Flat vs. recency-weighted ARI, frontier sweep (varying-density stream, $n=2000$, seed 7, $\lambda_{\text{eval}} = 0.99$, window = 100 points).

Model	Peak memory (KB)	ARI (flat)	ARI (recent)	Decayed purity
FixedGrid $n=2$	10.0	-0.001	0.000	0.542
FixedGrid $n=3$	13.1	0.001	0.000	0.542
FixedGrid $n=4$	17.7	0.001	-0.004	0.542
DBSTREAM	22.9	0.236	0.481	0.978
FixedGrid $n=6$	32.7	0.004	-0.008	0.542
DenStream	47.8	0.092	0.397	0.991
FixedGrid $n=8$	53.2	0.064	-0.010	0.542
FixedGrid $n=11$	96.2	0.029	0.717	0.976
CluStream	99.0	0.061	0.485	0.980
FixedGrid $n=16$	199.7	0.203	0.677	0.991
FixedGrid $n=22$	378.0	0.442	0.842	0.999
AdaptiveDStream	666.4	0.066	0.582	0.947
FixedGrid $n=32$	798.6	0.692	1.000	0.999
FixedGrid $n=45$	1581.4	0.822	0.978	0.996

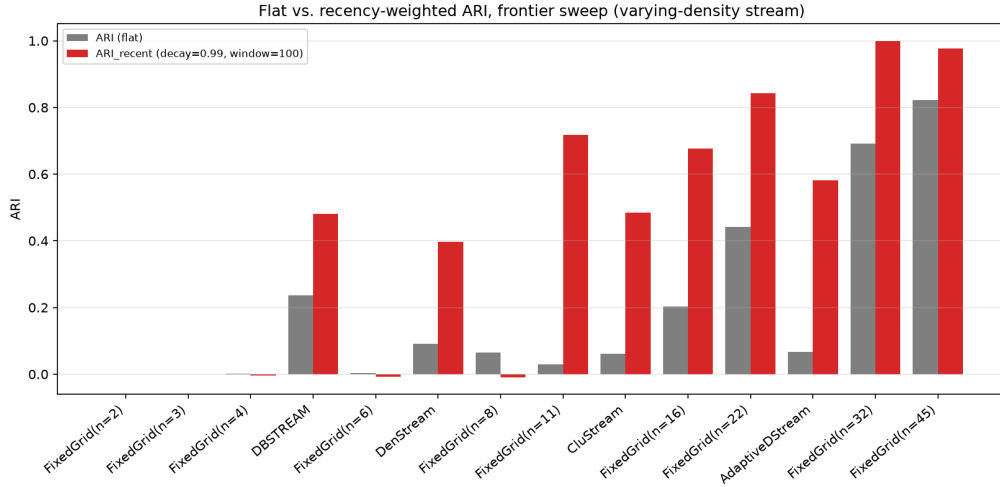


Figure 9: Flat vs. recency-weighted ARI, frontier sweep.

The same comparison plotted against memory, in the same form as the flat memory-accuracy frontier (Figure 1) rather than as a per-model bar chart, alongside it rather than in place of it:

Regime sweep: mixed, and in AdaptiveDStream’s favor more often than not. Figure 11 shows ARI (flat) $\rightarrow$ ARI (recent) for AdaptiveDStream across the five regimes of Table 5: **static_no_drift** 0.123 $\rightarrow$ 0.214, **spatial_drift** 0.030 $\rightarrow$ 0.026 (flat), **density_drift** 0.109 $\rightarrow$ 0.029 (worse), **two_gaussian_drift** 0.041 $\rightarrow$ 0.247 (a $6\times$ jump), **moons_drift** 0.004 $\rightarrow$ 0.040. Four of five regimes improve or hold. **density_drift** is the one exception, and it is plausibly a real effect rather than a metric artifact: that regime’s dense cluster is deliberately widening over time ($\sigma : 0.15 \rightarrow 0.15 \rightarrow 1.1$), so the most recent phase is also the hardest one to resolve — a lower **ari_recent** there is arguably measuring genuine difficulty, not a flaw in the lens.

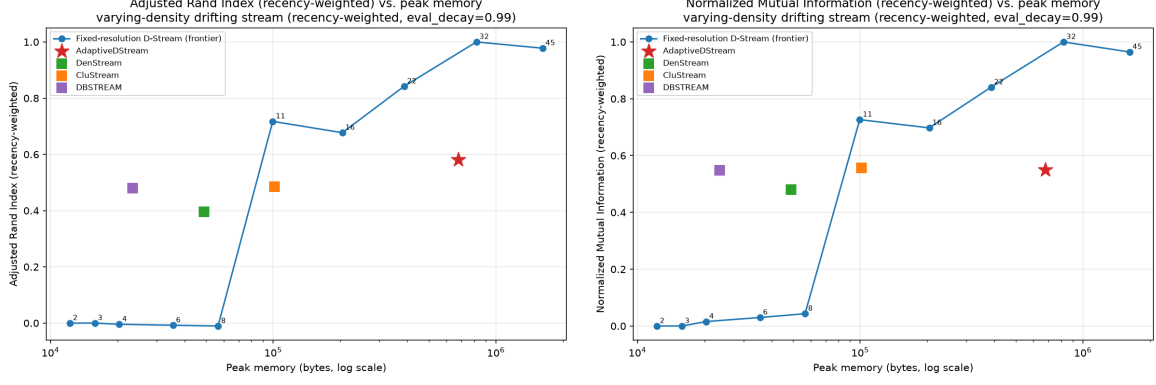


Figure 10: Recency-weighted ARI and NMI vs. peak memory, frontier sweep ($\lambda_{\text{eval}} = 0.99$).

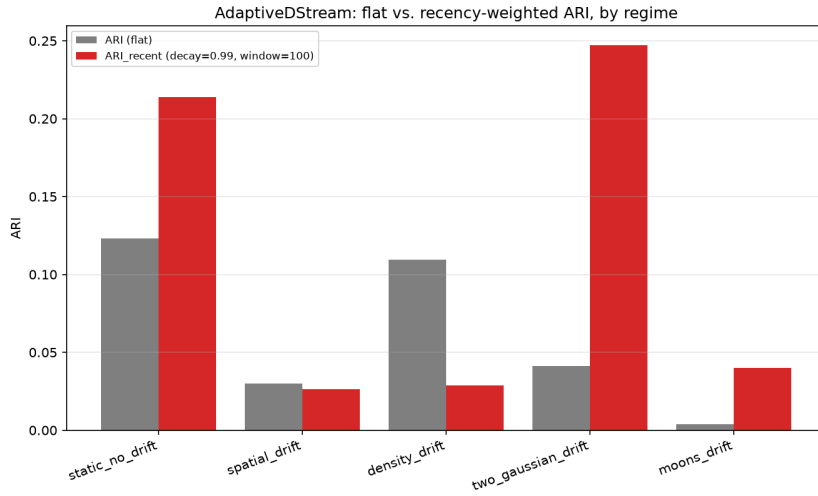


Figure 11: AdaptiveDStream: flat vs. recency-weighted ARI, by regime.

Split-strategy and dimensionality sweeps show the same broad pattern as the frontier sweep. Every split strategy improves substantially under the recent lens (equal_uniform $0.024 \rightarrow 0.527$, point_mass $0.017 \rightarrow 0.335$, moment_based $0.021 \rightarrow 0.300$), with the same relative ordering preserved as under the flat metric — recency-weighting does not change which strategy looks best, only by how much every strategy trails the fixed-grid frontier (Fixed-Grid $n=32$: $0.667 \rightarrow 0.903$). The dimension sweep likewise shows $\text{ARI}_{\text{recent}} \geq \text{ARI}$ for nearly every row across $d \in \{2, \dots, 20\}$, consistent with “early-stream underperformance while decayed mass accumulates” being a shared effect across models and dimensions, not something specific to AdaptiveDStream. One point worth noting: $d=10$ is the one dimension where AdaptiveDStream’s flat ARI (0.047) already edges out the fixed grid’s (0.030; Section 8.5 read this as plausibly small-sample noise), and ari_recent preserves the same ordering (0.129 vs. 0.082) rather than reversing it — mildly more supportive of a real effect than pure noise would predict, though still a single seed.

Recency-weighting is not a way to make the headline result disappear. The frontier-sweep conclusion (Section 8.3) holds under this lens too. But it materially changes the regime-sweep story: under a lens consistent with the method’s own design assumption — that recent data should matter more — AdaptiveDStream looks meaningfully better in most of the regimes tested. Both views belong in the record side by side, not one instead of the other; full numbers (including nmi_recent and the exact window size per run) are in `outputs/{frontier,regime_sweep,split_strategy,dimension_sweep}_results.json`.

8.10 Summary of Findings

The current prototype is not yet competitive with either the fixed-grid frontier or the imported published baselines, on ARI, on any of the streams evaluated, synthetic or real. Two of the three items added since the previous version of this evaluation, contraction and transitional-cell assignment, are measurable, unambiguous successes at what they targeted: peak memory and fraction unassigned both improved substantially and consistently across the frontier sweep, the regime sweep, and the dimensionality sweep. Neither improved ARI, and the split-strategy comparison (Section 8.4) shows the third addition, alternative split-mass redistribution, does not either.

A fourth addition, since the previous version of this evaluation, does address the dense/sparse threshold’s lack of volume normalization directly: `AdaptiveDStream(density_normalize=True)` rescales a cell’s decayed count by $\text{domain_volume}/\text{vol}(g)$ before the threshold comparison, exactly the density-versus-absolute-count fix Section 8.3 identifies as the cause. This was implemented and evaluated, and *reduces* ARI to chance level at every threshold tried, not the improvement the diagnosis predicted. The scale factor grows as $2^{d\cdot\text{level}}$, overshooting by orders of magnitude well before `max_depth` is reached and causing almost any cell that has split a few levels deep to read as dense regardless of whether that reflects real local density — a mechanism problem in the specific (undamped, uncapped) scaling used, not evidence the underlying diagnosis in Section 8.3 is wrong. This is reported here as a negative result rather than left untested: the direction motivated by Section 8.3 does not work as a simple linear rescaling, and a damped (e.g. $\sqrt{\cdot}$ or capped) variant is the natural next attempt, not yet tried.

Section 8.8 extends the evaluation to three real datasets and finds the same qualitative pattern as the synthetic streams: `AdaptiveDStream` is not competitive with the imported baselines, and a degeneracy check confirms, on real (not synthetic-noise) features, that the 2^d -split-rule identified in Section 8.5 is a hard architectural wall independent of any threshold fix — at `kd-cup99` `dim=12` with `max_cells=4000`, the model cannot perform a single split for the entire run. With the direct density-normalization fix now tried and found insufficient on its own, and the split-rule wall now confirmed on real data, we consider the per-axis split-rule redesign the higher-priority remaining item, ahead of further threshold-normalization variants.

The recency-weighted re-evaluation in Section 8.9 does not change this picture on the frontier sweep specifically — the fixed-grid frontier’s dominance survives a lens that weights recent points more, consistent with the method’s own decay assumption — but it does change the cross-regime picture materially: `AdaptiveDStream` improves under that lens in four of the five regimes tested, in one case (`two_gaussian_drift`) by more than $6\times$. All ARI numbers elsewhere in this section are the conventional, flat metric; Section 8.9 should be read alongside them, not as a correction superseding them.

We consider all of this an honest characterization of a first working implementation rather than a limitation of the underlying idea — the memory and unassigned-fraction results confirm that *refine-where-dense, merge-or-assign-where-not* behaves as designed.

9 Discussion

The proposed framework targets a central limitation of fixed-grid stream clustering algorithms: the inability to adapt resolution to local data complexity. The regime-sweep results in Table 5 support the motivating premise directly — no single fixed-grid resolution wins across the five regimes tested, since coarse cells that work when clusters are static or symmetric collapse once a tight cluster must be tracked through drift, and cells fine enough for that adversarial case are wasted, or actively harmful, on a stream with no density mismatch to justify them. That is exactly the failure mode a resolution that adapts to local structure is meant to avoid.

The current implementation of `AdaptiveDStream` does not yet realize that advantage on

ARI: it is dominated by the fixed-grid frontier, and by DBSTREAM, in every regime tested (Section 8.3). This is reported as the honest current state of the method rather than a result tuned to look favorable. Contraction and transitional-cell assignment (Section 4) were added specifically to make "refine where mass is dense, merge or assign where it is not" actually pay off in memory and in how many points get a usable label, and the results confirm that half of the design works: peak memory and fraction unassigned both improve substantially and consistently across every experiment in Section 8. What those two features cannot fix, because neither touches it, is the dense/sparse threshold itself — an absolute decayed count rather than a density (Section 8.3) — which is why ARI does not move with them, including on the no-drift static control, ruling out drift-tracking as the source of the remaining gap.

The use of time-decayed density remains appropriate for evolving streams in principle: older data gradually loses influence, letting the clustering structure track recent patterns, and sparse-cell pruning, now complemented by contraction, avoids retaining memory for inactive or over-refined regions. Realizing accuracy parity in practice is not as simple as the density-normalization fix in Section 8.3 alone: Section 8.8 reports that a direct implementation of it (rescaling by $2^{d_{\text{level}}}$) makes ARI worse, not better, so the diagnosis in Section 8.3 is correct but a linear rescaling overcorrects — a damped or capped variant remains untested and is the immediate next step, not a solved problem. Section 8.8 additionally identifies a second, independent architectural constraint: the full- 2^d -orthant split rule makes the model unable to split at all once 2^d exceeds the memory budget, which is a harder prerequisite than threshold calibration at realistic feature counts. Beyond both, refinement and merging thresholds must still be chosen carefully even once density-normalized: too aggressive, and memory grows quickly; too conservative, and clusters are represented too coarsely. Adaptive threshold selection remains an important direction for future work.

10 Conclusion

This paper proposed an adaptive multi-resolution grid-based framework for clustering massive data streams. The method extends fixed-resolution grid-based approaches by allowing grid cells to be dynamically refined, merged, and pruned according to local statistical summaries and memory constraints. Each cell stores compact information such as time-decayed density, center of mass, and variance. Clusters are extracted from connected dense cells using a density-based procedure, extended with a border-assignment rule for transitional cells.

The proposed approach is designed to improve the memory-accuracy trade-off of grid-based stream clustering methods and to better handle non-uniform, evolving data streams. An empirical comparison against a fixed-resolution frontier and three imported streaming-clustering baselines across five data regimes (Section 8.7 and Table 5), a dimensionality sweep (Section 8.5), a comparison of three split-mass-redistribution strategies (Section 8.4), and three real streaming-clustering benchmarks (Section 8.8) confirms the motivating premise — no single fixed grid resolution is competitive across all regimes — while also showing that contraction and transitional-cell assignment, though both measurably effective at the memory and unassigned-fraction problems they target, do not close the remaining ARI gap on their own. A direct density-normalization fix for the identified root cause (Section 8.3) was implemented and tested and did not close the gap either, and a degeneracy check on real data (Section 8.8) identifies a second, architectural cause — the full- 2^d -orthant split rule — as a hard prerequisite for reaching realistic feature counts, independent of threshold calibration. Future work includes a per-axis split rule, a better-calibrated density normalization, developing stronger theoretical guarantees, incorporating dimensionality reduction, improving concept drift detection, and extending the experimental comparison to additional regimes and real-world streams at larger scale.

Future Work

Several extensions can further improve the proposed framework. The first is the immediate, already-verified priority identified in Section 8.3; the two items after it are now partially addressed (contraction for the second, three redistribution strategies for the third) but not closed out, as discussed in Sections 8.3–8.4:

- **A per-axis split rule, replacing the full 2^d -orthant split:** Section 8.8’s degeneracy check confirms, on real feature spaces, that a cell can become architecturally unable to split at all once 2^d alone exceeds `max_cells` – independent of any refinement-criterion or threshold change. This is now the higher-priority remaining item: it gates whether the method can be evaluated at realistic feature counts (e.g. kddcup99’s 38 numeric columns) in the first place.
- **A better-calibrated density-normalized dense/sparse threshold:** the direct fix – replacing the absolute decayed-count comparison in $D(g, t) \geq \tau_{\text{dense/sparse}}$ with a comparison on density, $D(g, t)/\text{vol}(g)$ – was implemented and evaluated (Section 8.8’s Summary of Findings) and made ARI worse, not better, because the resulting scale factor ($2^{d \cdot \text{level}}$) grows too aggressively with split depth. A damped variant (e.g. $\sqrt{\cdot}$, or capped at a maximum multiplier) is the natural next attempt, to be evaluated before any threshold-normalization approach is adopted.
- **Extending contraction to reclaim parent storage more broadly:** Contraction (Section 4) already reclaims roughly 15% of total live cells in the measurement behind Section 8.3, but a cell whose children keep legitimately deviating from the uniform assumption never satisfies the merge criterion and keeps its stale parent resident. A more general reclamation strategy, not contingent on the merge criterion, is the natural next step.
- **Retuning `merge_threshold/merge_min_age` per regime:** both were fixed at one value across every experiment in Section 8, deliberately, to isolate the effect of adding contraction at all; a follow-up sweep over these two parameters is a natural extension once the threshold-normalization item above is addressed, since a resolution-aware dense/sparse rule will change what a ”stale” split looks like.
- **Streaming dimensionality reduction:** Random projections, incremental PCA, or streaming autoencoders can be used before grid construction to reduce the effect of high dimensionality.
- **Adaptive threshold learning:** The refinement, merging, and pruning thresholds can be adjusted automatically based on memory pressure, density variation, or clustering stability.
- **Explicit concept drift detection:** Drift detectors such as ADWIN, Page-Hinkley, DDM, or EDDM can be incorporated to trigger refinement or reorganization of the grid.
- **Sketch-based summaries:** Count-Min Sketch, reservoir sampling, or exponential histograms can be used to improve the accuracy of cell-level summaries.
- **Incremental connectivity maintenance:** Instead of recomputing clusters from scratch, dynamic graph connectivity or union-find structures can maintain clusters over dense cells incrementally.
- **Distributed and GPU implementations:** The grid update and clustering steps can be parallelized for high-throughput data streams.

A Derivations for the Weyl and Random-Projection Criteria

This appendix collects the derivations behind the two criteria of Section 4.4. All statements are under H_0 : $u \sim \text{Uniform}([0, 1]^d)$, with $u = (x - \ell(g))/h_{\ell(g)}$ the cell-local rescaling.

A.1 Weyl Criterion

Proposition 1 (Exact null moments). *For $k \in \mathbb{Z}^d \setminus \{0\}$, $\mathbb{E}[\varphi_k(u)] = 0$ and $\mathbb{E}[|\varphi_k(u)|^2] = 1$ exactly. For two distinct nonzero integer frequencies $k \neq \pm k'$, every second cross moment between $(\text{Re } \varphi_k, \text{Im } \varphi_k)$ and $(\text{Re } \varphi_{k'}, \text{Im } \varphi_{k'})$ is exactly zero.*

Proof. The characters $\{e^{2\pi i k \cdot u}\}_{k \in \mathbb{Z}^d}$ are an orthonormal basis of $L^2([0, 1]^d)$, so $\int_{[0, 1]^d} e^{2\pi i k \cdot u} du = \mathbb{1}_{\{k=0\}}$ for integer k . This gives $\mathbb{E}[\varphi_k(u)] = 0$ for $k \neq 0$ directly, and $\mathbb{E}[|\varphi_k(u)|^2] = \mathbb{E}[1] = 1$ since φ_k has unit modulus. For the cross moments, $\mathbb{E}[\varphi_k(u)\varphi_{k'}(u)] = \int e^{2\pi i(k-k') \cdot u} du = \mathbb{1}_{\{k=k'\}}$ and $\mathbb{E}[\varphi_k(u)\varphi_{k'}(u)] = \int e^{2\pi i(k+k') \cdot u} du = \mathbb{1}_{\{k=-k'\}}$; both vanish under $k \neq \pm k'$, and expanding into real and imaginary parts gives all four cross second moments equal to zero. $\square$

Proposition 1 gives $\text{Var}(\text{Re } \varphi_k) = \text{Var}(\text{Im } \varphi_k) = \frac{1}{2}$ with zero covariance, exactly, and zero cross-frequency covariance for the frequency set $\mathcal{K}$ (which is chosen with no two frequencies equal or opposite).

Theorem 1 (Asymptotic calibration). *Under H_0 , as $n \rightarrow \infty$, $T_{\text{Weyl}}(g) = \sum_{k \in \mathcal{K}} 2n |S_k|^2 \Rightarrow \chi_{2m}^2$, where $m = |\mathcal{K}|$.*

Proof. For each $k \in \mathcal{K}$, $\sqrt{n}(\text{Re } S_k, \text{Im } S_k)$ is a normalized sample mean of i.i.d. bounded terms with mean 0, covariance $\frac{1}{2}I_2$, and, by Proposition 1, zero cross-frequency covariance. The multivariate central limit theorem gives joint convergence of the stacked vector over all $k \in \mathcal{K}$ to $2m$ independent $\mathcal{N}(0, \frac{1}{2})$ coordinates. Then $2n|S_k|^2 = (\sqrt{2n} \text{Re } S_k)^2 + (\sqrt{2n} \text{Im } S_k)^2$ converges to a sum of two independent squared standard normals, i.e. χ_2^2 ; summing the m independent limits gives χ_{2m}^2 . $\square$

A.2 Random-Projection Criterion

Proposition 2 (Exact projected null moments). *For a fixed unit vector v , $Y = v \cdot u$ has $\mathbb{E}[Y] = \frac{1}{2} \sum_j v_j$ and $\text{Var}(Y) = \frac{1}{12} \|v\|_2^2 = \frac{1}{12}$, independent of v and d . Moreover $Y \in [a(v), b(v)]$ with $a(v) = \sum_j \min(v_j, 0)$, $b(v) = \sum_j \max(v_j, 0)$, and range $R(v) = b(v) - a(v) = \sum_j |v_j|$.*

Proof. $Y = \sum_j v_j u_j$ with the u_j i.i.d. Uniform $[0, 1]$, so $\mathbb{E}[Y] = \sum_j v_j \mathbb{E}[u_j] = \frac{1}{2} \sum_j v_j$ and $\text{Var}(Y) = \sum_j v_j^2 \text{Var}(u_j) = \frac{1}{12} \sum_j v_j^2 = \frac{1}{12} \|v\|_2^2$. Each $u_j \in [0, 1]$, so Y is maximized (minimized) by setting $u_j = 1$ where $v_j > 0$ and $u_j = 0$ where $v_j < 0$ (and vice versa), giving the stated $a(v), b(v)$ and $R(v) = \sum_j |v_j|$. $\square$

Theorem 2 (Hoeffding, 1963 [16]). *Let $Y_1, \dots, Y_n$ be i.i.d. with $Y_i \in [a, b]$ and mean μ . For any $t > 0$,*

$$\Pr(|\bar{Y}_n - \mu| \geq t) \leq 2 \exp\left(-\frac{2nt^2}{(b-a)^2}\right).$$

For each of the m random unit directions v_i , define the mean sub-test statistic on $Y = v_i \cdot u$ and the variance sub-test statistic on $Z = (Y - \frac{1}{2} \sum_j v_{i,j})^2$. By Proposition 2, $Y \in [a(v_i), b(v_i)]$ with $\mathbb{E}[Y] = \frac{1}{2} \sum_j v_{i,j}$, and $Z \in [0, R(v_i)^2]$ with $\mathbb{E}[Z] = \text{Var}(Y) = \frac{1}{12}$ (the sub-test centers Z at the exact null mean, not the sample mean, so $Z_1, \dots, Z_n$ remain i.i.d. under H_0 and Theorem 2 applies directly to $\bar{Z}_n$).

Corollary 1 (Finite-sample false-positive control). *Run each of the $2m$ elementary tests (a mean and a variance sub-test per direction) at level $\alpha/(2m)$, i.e. reject its null when the corresponding Hoeffding tail bound at the observed deviation falls below $\alpha/(2m)$, and refine g if any elementary test rejects. Then $\Pr(\text{refine } g \mid H_0) \leq \alpha$ for every finite n .*

Proof. By Theorem 2 applied to $\bar{Y}_n$ (mean sub-test) and to $\bar{Z}_n$ (variance sub-test), each elementary test has probability at most $\alpha/(2m)$ of rejecting under H_0 , at every n . The event “some elementary test rejects” is the union of $2m$ such events; the union bound gives total probability at most $2m \cdot \alpha/(2m) = \alpha$. $\square$

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